

PRANAV N J

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Education

Siddaganga Institute of Technology, BE in Information Science	Sept 2023 – July 2027
• CGPA: 9.38/10	
Excellent PU College, PCMB	May 2023
• 12th: 94.5%	
JVS Primary and High School,	May 2021
• 10th: 95.64%	

Skills

Languages: C++, C, Javascript

CS Fundamentals: Data Structures, DBMS, OOPs

Web Development: HTML, CSS, JavaScript, React.js, Node.js, Express.js, PostgreSQL, MongoDB

Tools: Git, GitHub, VS Code

Soft Skills: Communication, Teamwork, Public Speaking, Logical Reasoning

Projects

Phishing Detection System | MERN, Python, ML, FastAPI, REST API

- Built ML-powered phishing detection microservices (React, FastAPI, Express, MongoDB, Web Extension) that classify malicious URLs with **93.89% accuracy** and **29ms prediction latency**.
- Eliminated HTTPS protocol bias via dataset augmentation (27K→55K URLs) after evaluating 4 ML algorithms, reducing **false positive rate to 4.79%** and preventing production failure.
- Developed **59-feature Random Forest classifier** with GridSearchCV hyperparameter tuning and StratifiedKFold 5-fold cross-validation, improving **precision by 7.1%** and achieving **98.3% AUC-ROC**.

Suraksha | TypeScript, PostgreSQL, WebSocket, WebRTC, CoreNLP

- AI-powered women's safety platform built in TypeScript (React + Node + PostgreSQL) delivering **sub-2 sec multi-channel emergency alerts**, AI voice distress detection, and encrypted real-time coordination for families.
- Engineered AI emergency voice distress detection with a **3-stage pipeline** (AssemblyAI → CoreNLP → Llama 2), reaching **96% accuracy** and cutting **false positives by 35%** via weighted keywords plus rule-based fallback.
- Optimized 24/7 **real-time location tracking** with 15s GPS WebSocket beacons and **WebRTC live video streaming**, sustaining **97.8% multi-channel delivery** across 37+ active emergency sessions.

Server-Monitor | MySQL, Python, FastAPI, WebSocket, React, psutil

- A database-centric server monitoring system where MySQL triggers and stored procedures for instant, server health alerts, with a real-time **WebSocket pipeline** pushing **live CPU/RAM/Disk metrics** every 3 seconds.
- Automated multi-channel **SMS and email alerts** for critical server issues with SQL-enforced **5-minute cooldown**, centralized dispatch logging, and minimal backend complexity.
- Engineered database-driven **Recharts** visualizations for metric history, uptime tracking, and paginated logs improving **performance 75%** across time ranges.

Achievements

Technical Competitions: Selected participant for the **Smart India Hackathon 2025**, a premier national-level hackathon by the Government of India; Participant at **HackNite'25** organised by MIT Bengaluru; Participant at **National Space Hackathon** hosted by IIT Delhi.

Leadership and Service: Leveraged disciplined leadership as a Certified **National Cadet Corps (NCC)** Cadet and Event Management Team Member for the **Institution's Innovation Council (IIC-SET)** to drive organizational excellence.